

JKBOSE 2026

CLASS 11TH

Photosynthesis in Higher Plants

LECTURE-07

BOTANY

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Topics To Covered

1 Calvin Cycle

2 Photorespiration

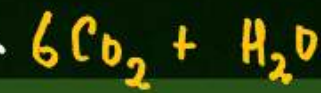
C_3 – Cycle or Calvin Cycle

- Discovered by Melvin Calvin by using ^{14}C radioactive isotope in Chlorella.
- First stable product formed is 3-phosphoglyceric acid (3-PGA) → It is a 3-carbon organic acid i.e., why the name C_3 -cycle.

• Examples of C_3 plants:

- Triticum (wheat),
- Oryza (rice),
- Bell pepper (capsicum),
- Lycopersicum (tomato),
- Solanum (potato).

Primary CO_2 acceptor in C_3 plants.



(RuBisCO)

6 x Ribulose 1,5 bisphosphate (5c)
(RuBP)

12 x 3-phosphoglyceric Acid (3c)
(3-PGA)

a.
Carboxylation

6 ADP $\leftarrow$
6 ATP

12 x ATP $\leftarrow$
12 x ADP

c.
Regeneration

b.
Reduction

12 x 1,3 bisphosphoglyceric Acid (3c)

12 P_i $\leftarrow$ 12 x $\text{NADPH} + \text{H}^+$
12 x NADP^+

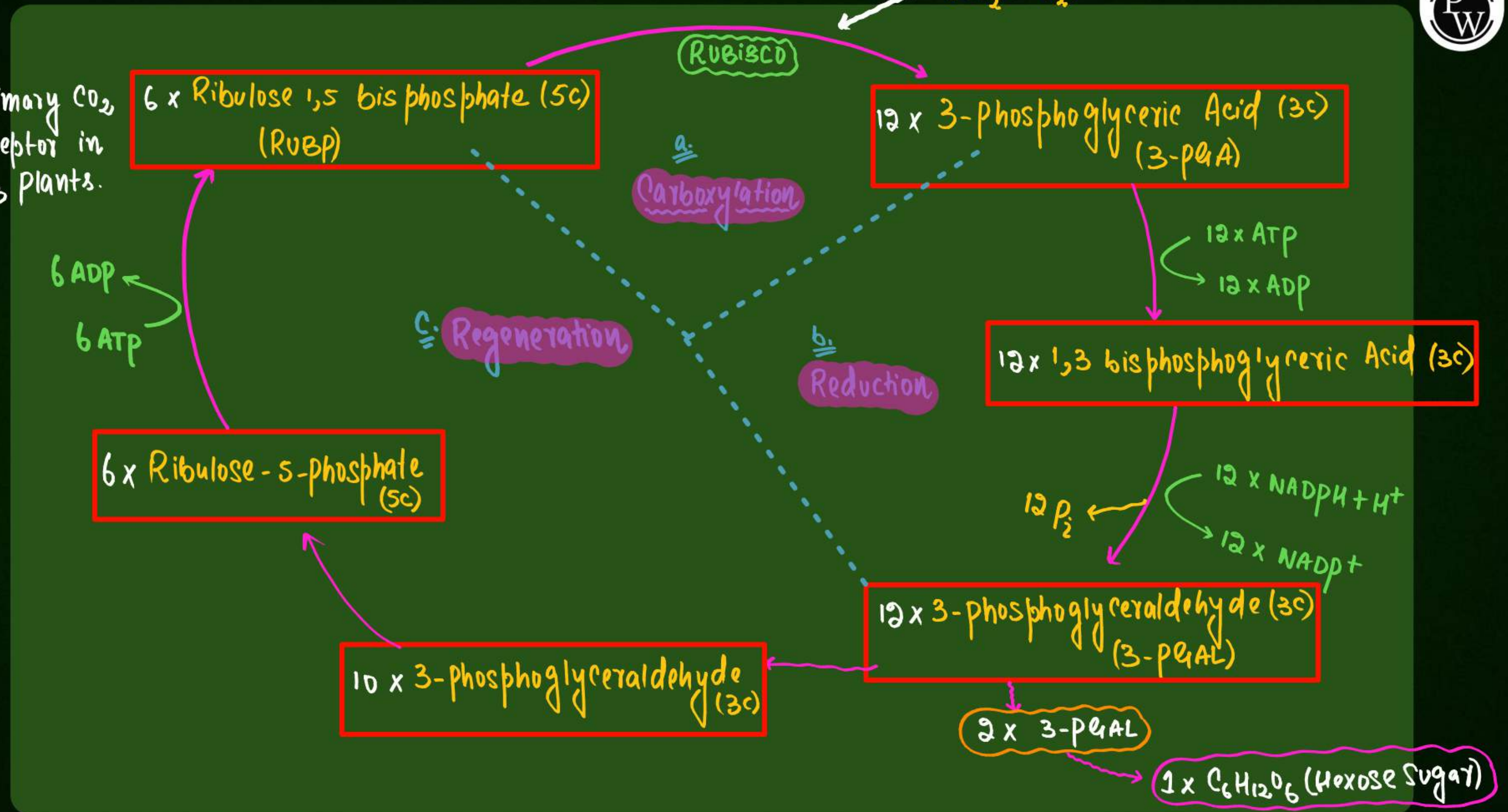
6 x Ribulose-5-phosphate
(5c)

12 x 3-phosphoglyceraldehyde (3c)
(3-PGAL)

10 x 3-phosphoglyceraldehyde
(3c)

2 x 3-PGAL

1 x $\text{C}_6\text{H}_{12}\text{O}_6$ (Hexose Sugar)



Rubisco



- ≡ Ribulose 1,5 biphosphate Carboxylase Oxygenase.
- ≡ It is a 5-c Compound.
- ≡ It is most abundant protein on planet Earth.
- ≡ It has dual activity
 - a. Carboxylase activity → fix CO_2 .
 - b. Oxygenase activity → fix O_2 .
- ≡ It shows carboxylase activity when → $\text{CO}_2 > \text{O}_2$ or $\text{CO}_2 = \text{O}_2$
- ≡ It shows oxygenase activity when → $\text{CO}_2 < \text{O}_2$

Important Points

- To fix 6 molecules of CO_2 $\longrightarrow$ Six turns of Calvin Cycle are required.
- To form 1 molecule of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) $\longrightarrow$ Six turns of Calvin cycle are required.
- For every CO_2 molecule $\longrightarrow$ 2 $\text{NADPH} + \text{H}^+$ and 3 ATP are required.
- To form 1 molecule of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), $\longrightarrow$ 12 $\text{NADPH} + \text{H}^+$ and 18 ATP are required.

For Single Turn of Calvin Cycle

Stages of C_3 -cycle	No. of ATP reqd.	No. of NADPH + H^+ reqd.
Carboxylation	X	X
Reduction	2 ATP	2 x NADPH + H^+
Regeneration	1 ATP	X

For 6 Turns of Calvin Cycle

IN	OUT
6 x CO_2	1 x $C_6H_{12}O_6$
18 x ATP	18 x ADP
12 x NADPH + H^+	12 x $NADP^+$

Photorespiration

Otto Warburg

Examined when

O_2 conc. ↑↑↑

CO_2 conc. ↓↓↓

Rate of photosynthesis decreases

Warburg Effect**

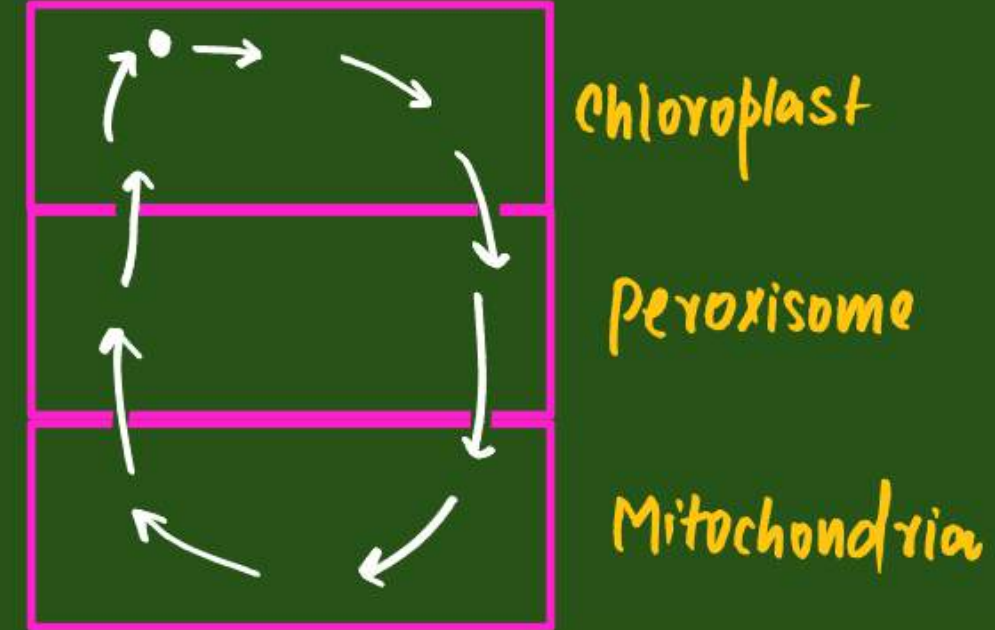
photorespiration.

Photorespiration

- Otto Warburg discovered with increase in O_2 Conc., the rate of photosynthesis decreases — He termed it Warburg Effect.
- The process of photorespiration was discovered by **Tid** and **Decker** in tobacco plant.
- It is also called **C_2 cycle** because first product formed is 2-C compound namely phosphoglycolate.
- It is AKA **glycolate cycle** or **glycolate metabolism**.

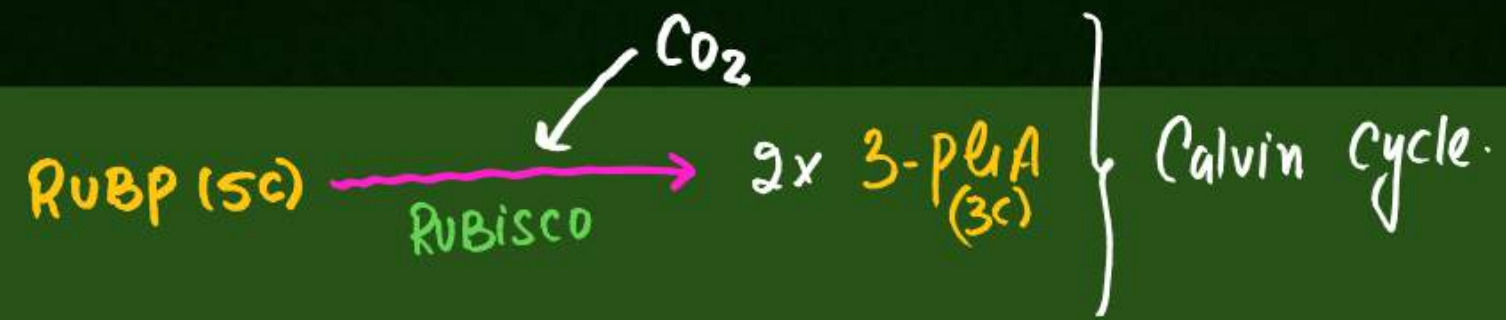
• SITE: 03 cell organelles of mesophyll

- (a) Chloroplast
- (b) peroxisome
- (c) Mitochondria



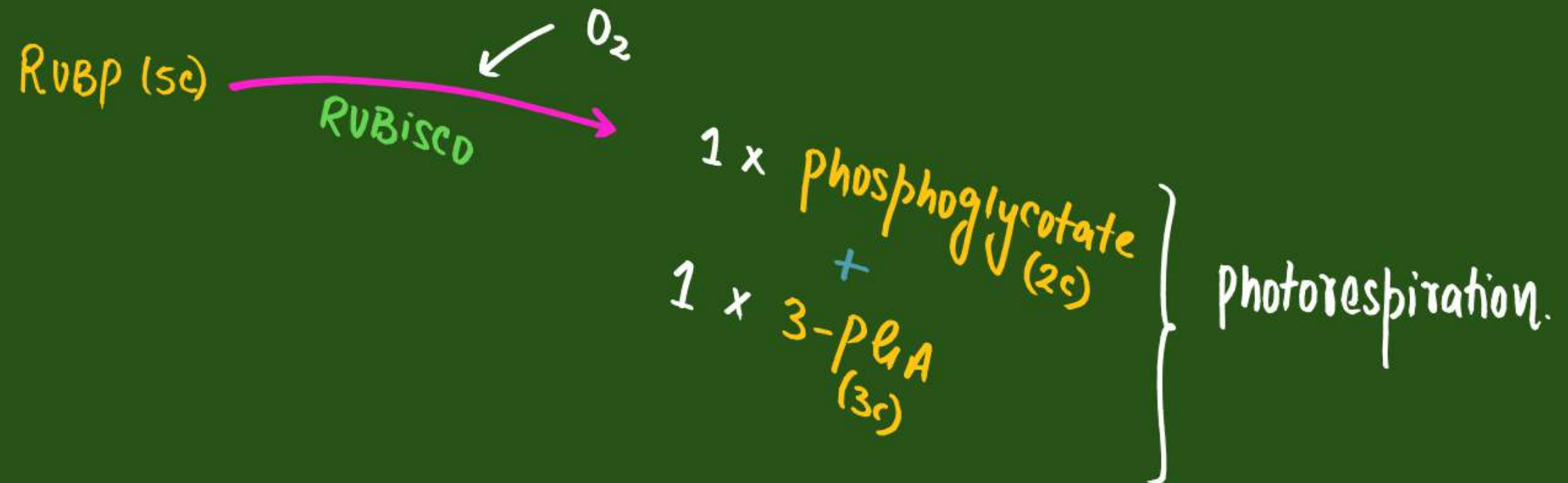
• Conditions under which photorespiration occurs:

- (1) $O_2 > CO_2$ OR $\frac{CO_2}{O_2} < 1$ OR $\frac{O_2}{CO_2} > 1$
- (2) High light intensity.



Note

- When RuBP combines with CO_2 , two molecules of 3-PGA (3C) are formed $\rightarrow$ (Calvin cycle).
- When RuBP combines with O_2 , one phosphoglycolate (2C) and one 3-PGA (3C) are formed $\rightarrow$ (photorespiration).



Important Points

→ Regarding Photorespiration.

Next lecture

QUESTIONS

How many molecules of ATP and NADPH are required for every molecule of CO₂ fixed in the Calvin cycle? (2024)

- A** 3 molecules of ATP and 3 molecules of NADPH
- B** 3 molecules of ATP and 2 molecules of NADPH
- C** 2 molecules of ATP and 3 molecules of NADPH
- D** 2 molecules of ATP and 2 molecules of NADPH

QUESTIONS

u/w

How many ATP and NADPH are required for the synthesis of one molecule of glucose during Calvin cycle? (2023)

- A** 18 ATP and 16 NADPH
- B** 12 ATP and 12 NADPH₂
- C** 18 ATP and 12 NADPH
- D** 12 ATP and 16 NADPH₂

QUESTIONS

μ/w

The primary acceptor, during CO_2 fixation in C_3 plants, is:

(1996)

- A** Phosphoenolpyruvate (PEP)
- B** Ribulose 1,5-biphosphate (RuBP)
- C** Phosphoglyceric acid (PGA)
- D** Ribulose monophosphate (RMP)

QUESTIONS

4/5

The carbon dioxide acceptor in Calvin cycle / C_3 plants is:

(1993)

- A** Phosphoenolpyruvate (PEP)
- B** Ribulose 1,5-biphosphate (RuBP)
- C** Phosphoglyceric acid (PGA)
- D** Ribulose monophosphate (RMP)

HOMework

03 Questions





Thank
You